

Vasantha G

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EDUCATION

- **Campbellsville University** Kentucky, USA
Master's in Information Technology Management; GPA: 4.0
- **Jawaharlal Nehru Technological University Kakinada** Kakinada, India
Bachelor of Technology in Information Technology; GPA: 9.07/10.0

SKILLS

Python, SQL, PySpark, Databricks, Kafka, Hadoop, Azure, MySQL, Oracle, Cassandra, Azure Data Factory, Synapse, Apache Airflow, ADLS, GitHub, Linux, Tableau, ETL

EXPERIENCE

- **TD Bank - Data Engineer** Jan 2025 - Present
Loan Repayment Behavior Analytics
 - Design and automate **Azure Data Factory** pipelines to ingest weekly loan data from JSON and Excel files into ADLS Gen2, ensuring timely and accurate data availability for **credit risk analysis**.
 - Clean, standardize, and enrich datasets using **PySpark** in **Azure Databricks** by applying delinquency rules, computing credit risk scores, and resolving schema inconsistencies.
 - Implement **automated validation** and **anomaly detection** workflows that reduce data errors by 85%, minimize manual interventions, and help maintain SLA compliance.
 - Deliver curated datasets to **Azure Synapse Analytics** and build **Tableau dashboards** to monitor delinquency trends, repayment behavior, and key regulatory reporting metrics.
- **IBM India Pvt. Ltd - Associate Data Engineer** Feb 2023 – Dec 2023
GMOT securities & insights
 - Built and automated **scalable data pipelines** to validate transaction legitimacy using **federal KPIs**, integrating data from **MySQL, REST APIs, Cassandra, and SFTP sources**. Automated ingestion into **Azure Data Lake Storage (ADLS) Gen2** using **Azure Data Factory** and implemented the **Medallion Architecture** (bronze, silver, gold) for structured data transformation.
 - Designed and optimized **ETL pipelines** with **PySpark** and **Azure Databricks**, transforming raw data into **Delta Lake tables** and improving **Spark SQL query performance by 30%**. Implemented cleansing and standardization logic in the silver layer to ensure **consistency in financial transaction data**.
 - Set up real-time alerting in **Azure Logic Apps**, which replaced manual monitoring and freed up bandwidth equivalent to two full-time employees.
 - Implemented **role-based access controls** in Azure Data Lake and Databricks to restrict sensitive data exposure and ensure **compliance with internal security policies**.
 - Troubleshoot and resolved production issues to maintain 99% pipeline uptime, while deploying workflows across **multiple regions** using **Azure CI/CD**. Partnered with **data and analytics teams** to model and warehouse clean, reliable data for forecasting and business insights.
- **Swiggy - Associate Data Engineer** Jan 2022 - Jan 2023
Functional and Add Analytics Engineering
 - **Built and optimized scalable ETL pipelines** using Azure Data Factory to ingest data from on-premises systems, internal Swiggy web applications, and MySQL into Azure Data Lake Storage Gen2.
 - Implemented an **automated data purging framework using Python** to optimize storage on ADLS Gen2 and **enhanced script performance** for large-scale data operations.
 - **Extracted and transformed data from Hive** using complex SQL logic to support analytics on ad placement performance.
 - Collaborated with DevOps and compliance teams to enable **audit logging** and secure data pipelines using **tokenized credentials** and **managed identities** in Azure.
 - Worked closely with the BI team to build **Tableau dashboards** that highlighted the most profitable ad placements — contributing to a 4% raise in ad revenue.
 - Implemented robust **data quality checks and anomaly detection** mechanisms to ensure accuracy and consistency of the data.
 - **Led the Hive table purging process** by coordinating with Swiggy stakeholders and ensuring **data compliance and relevance**.
- **Cyient - Data Engineer Intern** Mar 2021 - Aug 2021
Operational Data Integration Team
 - Designed and implemented robust **ETL pipelines** using **Azure Data Factory** to ingest and integrate diverse **operational datasets** from on-premises **ERP systems, IoT sensor streams, and relational databases** into **Azure Data Lake Storage Gen2**.
 - Utilized **PySpark** within **Azure Databricks** to perform complex **data cleansing, transformation, and aggregation** on large-scale **sensor and logistics data**, ensuring high **data quality** and reliability.
 - Developed and optimized **SQL queries and data models** to support interactive **reporting and visualization**, achieving a 30% improvement in **query performance**.

- Collaborated cross-functionally with **data analysts** and **supply chain stakeholders** to translate **business requirements** into scalable **data workflows**, enabling enhanced **inventory forecasting** and operational **decision-making**.
- **Computer Power group - Data Analyst Intern** *Jan 2020 - Jun 2020*
Customer Feedback Analytics
 - **Aggregated** customer feedback data from multiple sources (Excel files, SQL databases, emails) using **SQL** and **Python**, enabling a centralized view of client sentiments.
 - **Cleaned and transformed** raw text data by removing duplicates, standardizing formats, and categorizing feedback using keyword-based classification.
 - Developed **interactive Tableau dashboards** to track recurring complaints, sentiment trends, and issue resolution timelines, improving transparency for leadership.
 - Automated the **data refresh pipeline** using scheduled scripts in **Python** and **SQL**, ensuring the dashboard remained up to date with minimal manual effort.
 - Collaborated with cross-functional teams to **translate analytical findings into process improvements**, resulting in a **20% reduction in repeated complaints** and increased **customer satisfaction**.

PROJECTS

- **On-Prem to Cloud migration project** *Oct 2020 – Dec 2020*
 - Built scalable data pipelines in **Azure Data Factory** to convert **JSON** files into **CSV** format and store the output in organized containers.
 - Mounted **Azure Blob Storage** to **Databricks** and applied **PySpark** for cleaning tasks—handling null values, filtering unused fields, and standardizing formats.
 - Automated transformation logic to reduce manual data handling and accelerate reporting cycles.
 - Developed **reusable Databricks notebooks** to improve efficiency and consistency in future ETL workflows.
 - Orchestrated ingestion-to-delivery using Azure tools, ensuring seamless data flow from source to destination.
 - Created **Tableau dashboards** to visualize sales performance and business trends, enabling leadership to make faster, data-informed decisions.
 - Ensured **data integrity** by embedding validation checks across the pipeline lifecycle, improving downstream trust in analytics.
- **Integrated Healthcare Data Pipeline for Predictive Analytics Project** *Feb 2025 - May 2025*
 - Led the design and deployment of a **full-scale ETL pipelines** using **Azure Data Factory** to bring together data from **Electronic Medical Records (EMR)**, prescriptions, lab reports, and appointments, streamlining data access for the healthcare team.
 - Used **PySpark** in **Azure Databricks** to carefully **clean, join, and transform** data coming from a mix of sources—structured **SQL databases**, APIs, and semi-structured **CSV files**—to create a unified, accurate dataset.
 - Implemented **Slowly Changing Dimension (SCD) Type-2** logic to keep a complete and detailed history of patient records, supporting comprehensive **longitudinal analytics** over time.
 - Built robust **data quality checks** including **deduplication, null value handling, and schema validation** to ensure the data was reliable and ready for use.
 - Stored the cleaned and processed data in **Delta Lake** and **Azure Synapse Analytics**, making it easily accessible for **business intelligence** and **advanced predictive modeling**.
 - Optimized **PySpark** jobs which cut down pipeline processing time by 20%, helping the team meet their **SLAs** and deliver insights faster.
 - Developed intuitive **Tableau dashboards** that visualized important metrics like patient risk scores, re-admission chances, and treatment results - giving healthcare analysts actionable insights to improve care plans.